```sql
CREATE TABLE platform (
    platform_id    INT AUTO_INCREMENT COMMENT 'Surrogate primary key',
    platform_name  VARCHAR(120) NOT NULL,
    platform_type  ENUM('STREAMING','SOCIAL','DOWNLOAD','BROADCAST','OTHER') NOT NULL,
    country        CHAR(2)      NOT NULL COMMENT 'ISO 3166-1 alpha-2',
    contact_email  VARCHAR(254) NOT NULL,
    created_at     TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT pk_platform       PRIMARY KEY (platform_id),
    CONSTRAINT uq_platform_name  UNIQUE (platform_name),
    CONSTRAINT ck_platform_email CHECK (contact_email LIKE '%_@_%.__%')
) COMMENT 'Distribution platforms that licence content';


CREATE TABLE creator (
    creator_id          INT AUTO_INCREMENT,
    name                VARCHAR(150) NOT NULL,
    creator_type        ENUM('INDIVIDUAL','BAND','LABEL','STUDIO','ESTATE') NOT NULL,
    email               VARCHAR(254) NOT NULL,
    phone               VARCHAR(20)  DEFAULT NULL,
    country             CHAR(2)      NOT NULL,
    payout_currency     CHAR(3)      NOT NULL DEFAULT 'INR' COMMENT 'ISO 4217',
    bank_account_enc    VARBINARY(512) DEFAULT NULL COMMENT 'Encrypted at application layer',
    bank_account_last4  CHAR(4)      DEFAULT NULL,
    tax_id_enc          VARBINARY(512) DEFAULT NULL COMMENT 'Encrypted at application layer',
    created_at          TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,
CREATE TABLE content (
    content_id    INT AUTO_INCREMENT,
    title         VARCHAR(255) NOT NULL,
    content_type  ENUM('AUDIO','VIDEO','IMAGE','TEXT','PODCAST') NOT NULL,
    genre         VARCHAR(50)  DEFAULT NULL,
    release_date  DATE         DEFAULT NULL,
    duration_sec  INT          DEFAULT NULL,
    `language`    CHAR(3)      DEFAULT NULL COMMENT 'ISO 639-3',
    created_at    TIMESTAMP    NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT pk_content          PRIMARY KEY (content_id),
    CONSTRAINT ck_content_duration CHECK (duration_sec IS NULL OR duration_sec > 0)
) COMMENT 'Licensable works';

CREATE TABLE ownership (
    creator_id           INT          NOT NULL,
    content_id           INT          NOT NULL,
    ownership_percentage DECIMAL(5,2) NOT NULL,
    `role`               ENUM('PRIMARY_ARTIST','FEATURED_ARTIST','COMPOSER',
                              'LYRICIST','PRODUCER','PUBLISHER','DIRECTOR') NOT NULL,

    CONSTRAINT pk_ownership   PRIMARY KEY (creator_id, content_id),
    CONSTRAINT fk_own_creator FOREIGN KEY (creator_id) REFERENCES creator(creator_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT fk_own_content FOREIGN KEY (content_id) REFERENCES content(content_id)
        ON DELETE CASCADE ON UPDATE CASCADE,
    CONSTRAINT ck_own_pct CHECK (ownership_percentage > 0 AND ownership_percentage <= 100)
) COMMENT 'Split of rights in a work between creators';
```

```sql
CREATE TABLE license (
    license_id    INT AUTO_INCREMENT,
    platform_id   INT           NOT NULL,
    content_id    INT           NOT NULL,
    license_type  ENUM('EXCLUSIVE','NON_EXCLUSIVE','SUBLICENSABLE') NOT NULL,
    start_date    DATE          NOT NULL,
    end_date      DATE          DEFAULT NULL COMMENT 'NULL means perpetual',
    royalty_rate  DECIMAL(6,4) NOT NULL COMMENT 'Fraction, 0.0000 to 1.0000',
    currency      CHAR(3)       NOT NULL,
    status        ENUM('DRAFT','ACTIVE','SUSPENDED','EXPIRED','TERMINATED')
                                NOT NULL DEFAULT 'DRAFT',
    created_at    TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT pk_license      PRIMARY KEY (license_id),
    CONSTRAINT fk_lic_platform FOREIGN KEY (platform_id) REFERENCES platform(platform_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT fk_lic_content  FOREIGN KEY (content_id) REFERENCES content(content_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT ck_lic_rate  CHECK (royalty_rate >= 0 AND royalty_rate <= 1),
    CONSTRAINT ck_lic_dates CHECK (end_date IS NULL OR end_date > start_date)
) COMMENT 'Grant of rights in one work to one platform';
```

```sql
CREATE TABLE license_territory (
    license_id INT        NOT NULL,
    territory  VARCHAR(3) NOT NULL COMMENT 'ISO alpha-2, or WW for worldwide',

    CONSTRAINT pk_lic_terr PRIMARY KEY (license_id, territory),
    CONSTRAINT fk_lic_terr FOREIGN KEY (license_id) REFERENCES license(license_id)
        ON DELETE CASCADE ON UPDATE CASCADE
) COMMENT 'Territories covered by a licence (1NF split)';


CREATE TABLE usage_record (
    usage_id         INT AUTO_INCREMENT,
    license_id       INT           NOT NULL,
    usage_date       DATE          NOT NULL,
    usage_type       ENUM('STREAM','DOWNLOAD','VIEW','BROADCAST','SYNC') NOT NULL,
    quantity         BIGINT        NOT NULL,
    gross_revenue    DECIMAL(14,2) NOT NULL,
    currency         CHAR(3)       NOT NULL,
    source_batch_ref VARCHAR(100)  NOT NULL COMMENT 'Report file identifier',
    ingested_at      TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT pk_usage         PRIMARY KEY (usage_id),
    CONSTRAINT fk_usage_license FOREIGN KEY (license_id) REFERENCES license(license_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT uq_usage_batch   UNIQUE (license_id, usage_date, usage_type, source_batch_ref),
    CONSTRAINT ck_usage_qty     CHECK (quantity > 0),
    CONSTRAINT ck_usage_rev     CHECK (gross_revenue >= 0)
) COMMENT 'Consumption reported by a platform against a licence';
```

```sql
CREATE TABLE payment (
    payment_id      INT AUTO_INCREMENT,
    creator_id      INT           NOT NULL,
    payment_date    DATE          NOT NULL,
    amount          DECIMAL(14,2) NOT NULL,
    currency        CHAR(3)       NOT NULL,
    method          ENUM('BANK_TRANSFER','UPI','PAYPAL','WIRE','CHEQUE') NOT NULL,
    status          ENUM('PENDING','PROCESSING','COMPLETED','FAILED','REVERSED')
                                  NOT NULL DEFAULT 'PENDING',
    transaction_ref VARCHAR(100)  DEFAULT NULL,

    CONSTRAINT pk_payment     PRIMARY KEY (payment_id),
    CONSTRAINT fk_pay_creator FOREIGN KEY (creator_id) REFERENCES creator(creator_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT uq_pay_txn     UNIQUE (transaction_ref),
    CONSTRAINT ck_pay_amt     CHECK (amount > 0)
) COMMENT 'Disbursement to a creator, settling one or more royalty lines';
```

```sql
CREATE TABLE royalty (
    royalty_id                  INT AUTO_INCREMENT,
    usage_id                    INT           NOT NULL,
    creator_id                  INT           NOT NULL,
    payment_id                  INT           DEFAULT NULL,
    calculated_amount           DECIMAL(14,2) NOT NULL,
    currency                    CHAR(3)       NOT NULL,
    rate_applied                DECIMAL(6,4)  NOT NULL COMMENT 'Licence rate at calculation time',
    ownership_percentage_applied DECIMAL(5,2)  NOT NULL COMMENT 'Split at calculation time',
    calculation_date            DATE          NOT NULL,
    period_start                DATE          NOT NULL,
    period_end                  DATE          NOT NULL,

    CONSTRAINT pk_royalty      PRIMARY KEY (royalty_id),
    CONSTRAINT fk_roy_usage    FOREIGN KEY (usage_id) REFERENCES usage_record(usage_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT fk_roy_creator  FOREIGN KEY (creator_id) REFERENCES creator(creator_id)
        ON DELETE RESTRICT ON UPDATE CASCADE,
    CONSTRAINT fk_roy_payment  FOREIGN KEY (payment_id) REFERENCES payment(payment_id)
        ON DELETE SET NULL ON UPDATE CASCADE,
    CONSTRAINT uq_royalty_line UNIQUE (usage_id, creator_id),
    CONSTRAINT ck_roy_amt      CHECK (calculated_amount >= 0),
    CONSTRAINT ck_roy_rate     CHECK (rate_applied >= 0 AND rate_applied <= 1),
    CONSTRAINT ck_roy_pct      CHECK (ownership_percentage_applied > 0
                                  AND ownership_percentage_applied <= 100),
    CONSTRAINT ck_roy_period   CHECK (period_end >= period_start)
) COMMENT 'Calculated earning of one creator from one usage record';
```

OUTPUT

```
+-----------------------------------+
| Tables_in_sandbox_db |
+-----------------------------------+
| content                           |
| creator                           |
| license                           |
| license_territory                 |
| ownership                         |
| payment                           |
| platform                          |
| royalty                           |
| usage_record                      |
+-----------------------------------+
```

```
+------------------------------+---------------+------+-----+---------+----------------+
| Field                        | Type          | Null | Key | Default | Extra          |
+------------------------------+---------------+------+-----+---------+----------------+
| royalty_id                   | int           | NO   | PRI | NULL    | auto_increment |
| usage_id                     | int           | NO   | MUL | NULL    |                |
| creator_id                   | int           | NO   | MUL | NULL    |                |
| payment_id                   | int           | YES  | MUL | NULL    |                |
| calculated_amount            | decimal(14,2) | NO   |     | NULL    |                |
| currency                     | char(3)       | NO   |     | NULL    |                |
| rate_applied                 | decimal(6,4)  | NO   |     | NULL    |                |
| ownership_percentage_applied | decimal(5,2)  | NO   |     | NULL    |                |
| calculation_date             | date          | NO   |     | NULL    |                |
| period_start                 | date          | NO   |     | NULL    |                |
| period_end                   | date          | NO   |     | NULL    |                |
+------------------------------+---------------+------+-----+---------+----------------+
```

```
+---------+-----------------------------------------------------------------------------------------------------------
| Table   | Create Table
+---------+-----------------------------------------------------------------------------------------------------------
| royalty | CREATE TABLE `royalty` (
  `royalty_id` int NOT NULL AUTO_INCREMENT,
  `usage_id` int NOT NULL,
  `creator_id` int NOT NULL,
  `payment_id` int DEFAULT NULL,
  `calculated_amount` decimal(14,2) NOT NULL,
  `currency` char(3) NOT NULL,
  `rate_applied` decimal(6,4) NOT NULL COMMENT 'Licence rate at calculation time',
  `ownership_percentage_applied` decimal(5,2) NOT NULL COMMENT 'Split at calculation time',
  `calculation_date` date NOT NULL,
  `period_start` date NOT NULL,
  `period_end` date NOT NULL,
  PRIMARY KEY (`royalty_id`),
  UNIQUE KEY `uq_royalty_line` (`usage_id`,`creator_id`),
  KEY `fk_roy_creator` (`creator_id`),
  KEY `fk_roy_payment` (`payment_id`),

  UNIQUE KEY `uq_royalty_line` (`usage_id`,`creator_id`),
  KEY `fk_roy_creator` (`creator_id`),
  KEY `fk_roy_payment` (`payment_id`),
  CONSTRAINT `fk_roy_creator` FOREIGN KEY (`creator_id`) REFERENCES `creator` (`creator_id`) ON DELETE RESTRICT ON UPDATE CASCADE
  CONSTRAINT `fk_roy_payment` FOREIGN KEY (`payment_id`) REFERENCES `payment` (`payment_id`) ON DELETE SET NULL ON UPDATE CASCADE
  CONSTRAINT `fk_roy_usage` FOREIGN KEY (`usage_id`) REFERENCES `usage_record` (`usage_id`) ON DELETE RESTRICT ON UPDATE CASCADE,
  CONSTRAINT `ck_roy_amt` CHECK ((`calculated_amount` >= 0)),
  CONSTRAINT `ck_roy_pct` CHECK (((`ownership_percentage_applied` > 0) and (`ownership_percentage_applied` <= 100))),
  CONSTRAINT `ck_roy_period` CHECK ((`period_end` >= `period_start`)),
  CONSTRAINT `ck_roy_rate` CHECK (((`rate_applied` >= 0) and (`rate_applied` <= 1)))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci COMMENT='Calculated earning of one creator from one usage reco
```